LandTrace

Automated Land Record Fraud Detection & Title Chain Integrity System

LandTech 2026

Problem P3 • Team ID: LA-73

1. Selected Problem Statement

P3: Fraud & Double-Selling Detection

LandTrace addresses land title fraud, unauthorized resale, and cross-registry record contradictions in Bangladesh by implementing an automated Ownership Chain Contradiction Detection Engine capable of analyzing historical custody chains and evaluating ongoing transaction proposals on demand.

2. Bangladesh Context & Administrative Background

Land governance in Bangladesh operates through a dual-channel administrative architecture divided across two separate ministries:

- **Deed Registration:** Administered by the Directorate of Registration under the Ministry of Law, Justice and Parliamentary Affairs through Upazila Sub-Registry offices, governing the execution and archiving of registered deeds (দলিল).
- **Record of Rights & Mutation (নামজারি / খতিয়ান):** Administered by Upazila Land Offices under the Ministry of Land, led by Assistant Commissioners (Land), governing cadastral survey records (CS, SA, RS, BS/City Survey) and mutation khatians.

Because Sub-Registry deed archives and AC (Land) mutation records historically functioned in administrative silos without automated cross-verification, opportunistic syndicates exploit procedural lag between deed registration and mutation approval to execute unauthorized transactions.

3. Problem Analysis

The absence of automated title custody cross-verification enables recurring fraud patterns in land transfers:

- **Double Selling (দ্বৈত বিক্রয়):** Predatory conveyance of previously alienated land to secondary buyers before the primary buyer completes the mutation process.
- **Ownership Chain Breaks (ধারাবাহিকতা বিচ্ছিন্ন):** Deeds executed by unverified or fictitious grantors who possess no legitimate predecessor title in the recorded chain of custody.
- **Area Inflation & Mismatches (পরিমাপের অসঙ্গতি):** Transfer deeds stipulating acreage that exceeds the parent cadastral survey khatian boundary or cumulative plot shares.
- **Deed–Mutation Discrepancies (দলিল-নামজারি অমিল):** Mutations sanctioned for acreage or applicants inconsistent with underlying registered deeds.
- **Judicial Backlog:** Land title contradictions and fraudulent conveyances constitute a predominant driver of protracted civil litigation and tribunal case backlogs across Bangladesh.

4. Proposed Solution: LandTrace

LandTrace is an automated **Decision-Support System (DSS)** designed for Sub-Registrars and Assistant Commissioners (Land). Instead of evaluating land records as disconnected paper entries, LandTrace models land holdings as directed title custody graphs, evaluating mathematical, spatial, and chronological integrity across five core fraud vectors on demand.

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Land Records (Deeds & Khatians) ➔ Data Normalization ➔ Directed Ownership Chain Engine
     ➔ Five-Vector Contradiction Audit ➔ Risk Scoring (0–100) ➔ Officer Decision Support
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Decision-Support Distinction: LandTrace equips land officers with structured forensic discrepancy findings; it does not adjudicate legal ownership or replace judicial authority.

LandTrace — Technical Architecture

Modular Full-Stack Architecture & Analytical Capabilities

P3 Concept Note

5. System Architecture

LandTrace operates as a modular full-stack web application with decoupled engine services, deployed on serverless cloud infrastructure:

- **Presentation Layer:** Bilingual administrative portal (বাংলা default / English) built with Next.js 16 App Router and Tailwind CSS v4, optimized for officer workflows.
- **Application & Logic Layer:** Server Components and route handlers executing a pure TypeScript topological contradiction engine with sub-second on-demand evaluation.
- **Data Persistence Layer:** Neon Serverless PostgreSQL 16 with SSL pooling, managed by Prisma 6 ORM for structured parcel, deed, mutation, and audit records.
- **Deployment Pipeline:** Continuous integration and deployment via GitHub and Vercel with automated test gates.

6. Technology Stack

Layer	Technologies	Role in Architecture
Frontend	Next.js 16, React 19, Tailwind CSS v4	Bilingual officer portal, interactive timelines, printable dossiers
Logic & Engine	TypeScript 5, Node Test Runner	Topological custody graph analysis and contradiction algorithms
Database	PostgreSQL 16 on Neon, Prisma 6 ORM	Cadastral parcels, deeds, mutations, and persistent audit logs
Hosting & CI/CD	Vercel Serverless, GitHub Actions	Automated zero-downtime production deployment on git push

7. Key System Features

- **Bilingual Administrative Portal:** Default বাংলা language aligned with official government land terminology (খতিয়ান, দাগ, মৌজা, নামজারি) with instant English toggle.
- **Interactive Title Timeline:** Chronological visual mapping of transfers from root survey grant down to current claimant with verification badges.
- **Pre-Registration Risk Check:** Interactive decision-support simulation tool allowing Sub-Registrars to audit proposed deeds before execution.
- **Forensic Evidentiary Dossier:** Structured discrepancy breakdown citing conflicting deed serials, parties, execution dates, and decimal variances.
- **Persistent Officer Action Trail:** Action logging tracking record reviews, field inquiry dispatches, and administrative inspection notes.

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Officer Web App (Bilingual: বাংলা | English)
     ↓
Next.js App Router API & Server Components
     ↓
Ownership Chain Contradiction Engine (Pure TypeScript)
     ↓
PostgreSQL on Neon Serverless (Prisma 6 Client)
     ↓
Calibrated Risk Score (0–100) + Forensic Evidence Dossier
```

LandTrace — Five Detection Capabilities

Algorithmic Rules, Mathematical Conditions & Administrative Value

P3 Concept Note

8. Five Core Detection Capabilities

Capability	Detection Logic & Mathematical Trigger	Administrative Impact
1. Double Selling (দ্বৈত বিক্রয়)	Flags instances where Seller S transfers land to Buyer A, and subsequently executes deed to Buyer B without documented intervening re-acquisition: `Date(Sale B) > Date(Sale A) ∧ Title(S) == 0`	Prevents unauthorized secondary conveyances of sold parcels before mutation.
2. Chain Break (ধারাবাহিকতা বিচ্ছিন্ন)	Flags deeds where Grantor G in deed N does not appear as a legitimate Grantee in preceding deed N-1, parent khatian, or inheritance record: `G ∉ ValidGranteeSet(N-1)`	Stops fraudulent conveyances executed by unverified or phantom grantors.
3. Area Mismatch (পরিমাপের গরমিল)	Flags conveyances where deed acreage exceeds the parent cadastral survey khatian acreage or cumulative plot shares: `Area(Deed) > Area(Khatian)`	Eliminates artificial acreage inflation and boundary encroachments.
4. Deed–Mutation Mismatch (দলিল-নামজারি অমিল)	Cross-validates AC-Land namjari records against sub-registry deeds for area discrepancies (`|Area(Mut) - Area(Deed)| > 0.05`) or applicants without registered deeds.	Ensures mutated records strictly conform to legitimate registered acquisition deeds.
5. Wrong Timing / Rapid Flip (সময়ানুক্রমিক অসঙ্গতি)	Detects anachronistic deeds (`Date(Sale) < Date(Acquisition)`) or rapid speculative flipping (`Interval < 7 days`) without interim namjari clearance.	Flags backdated fraudulent deeds and rapid speculative syndicate activity.

9. Prototype Implementation vs. Future Integration

Current Implementation (Round 1 Prototype): LandTrace currently operates as an independent decision-support prototype evaluating deterministic synthetic datasets across 7 benchmark parcels. It does not connect to live government systems, does not scrape public websites, and stores no real citizen records or PII.

Benchmark Dataset Coverage (Deterministic Synthetic Data):

- `PLOT-DH-1001` (Tejgaon, Dhaka): Fully verified 100% clean title custody chain (0 anomalies, CLEAN status).
- `PLOT-DH-2045` (Gulshan, Dhaka): Critical double-selling anomaly (unauthorized resale to second buyer).
- `PLOT-SYL-5501` (Sylhet Sadar): Critical chain break anomaly (phantom grantor absent from chain of title).
- `PLOT-CTG-3301` (Pahartali, Chattogram): High area mismatch anomaly (transferred acreage exceeds khatian).
- `PLOT-RAJ-4402` (Boalia, Rajshahi): High deed-mutation discrepancy (mutated area exceeds registered deed).
- `PLOT-KHL-6610` (Khalishpur, Khulna): Medium timing anomaly (rapid 48-hour resale without namjari).
- `PLOT-BAR-7703` (Barishal Sadar): Critical multi-contradiction composite benchmark (simultaneous violations).

LandTrace — Implementation & Governance

Roadmap, Integration Architecture & Team Information

P3 Concept Note

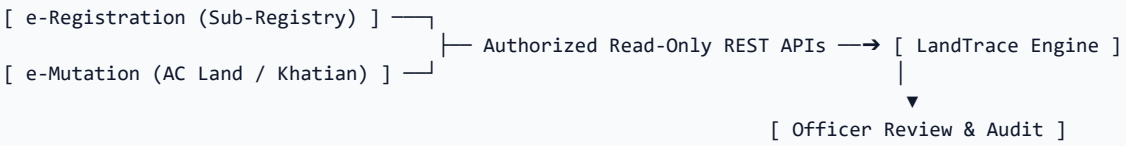
10. Countrywide Implementation Roadmap

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Phase 1 (Pilot – Months 1–4):
  Deploy in 2 Upazilas (e.g., Tejgaon, Dhaka and Sylhet Sadar) in parallel with e-Mutation and e-Registration.
Phase 2 (Divisional Rollout – Months 5–10):
  Integrate LandTrace API into divisional Land Commissionerates; train AC (Land), Sub-Registrars, and Kanungos.
Phase 3 (Nationwide Deployment – Months 11–18):
  Full nationwide rollout across all 64 districts with centralized Ministry of Land dashboards.
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11. Legal & Regulatory Boundary

Decision-Support Notice: LandTrace is designed strictly as an analytical decision-support tool for land officers. It does not issue judicial decrees, adjudicate legal ownership, or replace the statutory authority of Land Tribunals, Sub-Registrars, and Assistant Commissioners (Land) under the State Acquisition and Tenancy Act 1950 or Registration Act 1908.

12. Proposed Future Government Systems Integration Plan



Note: Illustrates the proposed production target architecture upon formal government onboarding, distinct from the current synthetic prototype.

13. Team & Mentor Details

Competition & Team	**LandTech 2026** | Problem P3: Fraud & Double-Selling Detection | **Team ID: LA-73**
Team Members	**Rafiul Kabir** (Team Leader) — Department of Software Engineering, Student ID: 253-35-162 **Mst Maimuna Mim** (Team Member) — Department of Software Engineering, Student ID: 253-35-103
Faculty Mentor	**Ms. Subarna Akter Liza** — Lecturer, Department of Computer Science & Engineering (CSE) Institutional Email: subarna.cse@diu.edu.bd • *Assigned by Organizing Committee*
Industry Support Mentor	**MD. Tahamid Hassan Sadib** — CSE-TECH Contact / Institutional ID: 251-15-084@diu.edu.bd • *Assigned by Organizing Committee*
Repository & Deployment	Live Prototype: **https://landtrace.vercel.app** GitHub Repository: https://github.com/RafiulKabir0/landtrace